

You are a data analyst working with an EdTech startup that wants to grow its offerings in recorded lectures. The company has collected data from various EdTech websites but needs your expertise to make sense of it. Your task is to clean and analyse this data to uncover valuable insights. To ensure the startup can effectively leverage this information, you will create a dashboard that presents the following insights. Your analysis will help the company identify areas for improvement and opportunities for growth in their recorded lecture services. The most important emphasis should be category-wise.

1. Calculate the total number of courses available in the dataset to establish the overall size of the catalog being analysed.
2. Calculate the average course rating across the dataset to provide a baseline understanding of overall content quality.
3. Calculate the total and average number of views across all courses to understand the overall scale of viewer engagement.
4. Examine the distribution of course types across categories to uncover trends and insights, enabling the client to strategically determine which course types to launch in specific categories for maximum impact and alignment with learner demand, also count the number of courses by category and sub-category.
5. Calculate the average number of views for each category, sub-category, and language to provide insights into viewer engagement patterns and inform strategic content development.
6. What is the distribution of various languages in which a particular course is created?
7. Determine the language preferences for each category based on viewer preferences, so that clients can optimise course accessibility and better align content with audience demand. Clients only want to analyse this data for the top 5 categories based on user preferences.

8. Investigate the relationship between the availability of subtitles and the number of views for courses to determine how subtitle options may impact viewer engagement and accessibility.
9. Identify the top three instructors for each category and subcategory based on ratings to highlight educators who consistently deliver high-quality content and effectively engage learners so that your client can approach them to make content for them and make this visual as static.
10. Examine the relationship between course duration and the number of views to understand how the length of a course may influence viewer engagement and preferences for each category and sub-category. If course duration is a month (in each month only 60 hours of content) and for flexible schedules make the timing 200 hours.